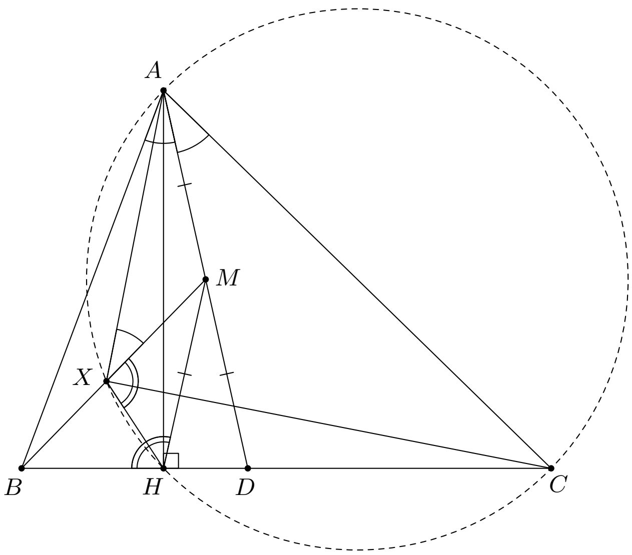
Problem 65. Let ABC be an acute-angled triangle with $AB < AC$. Let D be the foot of the internal angle bisector from A to BC and let M be the midpoint of the line segment AD. Finally, let X be a point on the line segment BM such that $\angle MXA = \angle DAC$. Show that the lines AX and XC are perpendicular to each other.

Solution 1.



Since $\angle MXA = \angle DAC = \angle MAB$, we have $\triangle MXA \sim \triangle MAB$. Therefore

$$\frac{XM}{AM} = \frac{AM}{BM}.$$

Let H be the foot of the perpendicular line from A to BC and so H lies between B and D (because $AB < AC$). Since M is the midpoint of the hypotenuse of the right triangle AHD, we have that $AM = HM$. Hence

$$\frac{XM}{HM} = \frac{HM}{BM},$$

and so $\triangle HXM \sim \triangle BHM$. Consequently,

$$\angle HXM = \angle BHM = 180^\circ - \angle MHD = 180^\circ - \angle MDH = 180^\circ - \left(\frac{1}{2}\angle BAC + \angle ACB\right),$$

where in the third equality we used the fact that $\triangle MHD$ is isosceles, since $\triangle AHD$ is a right angle and M is the midpoint of the hypotenuse. Hence

$$\begin{aligned}\angle AXH + \angle ACH &= (\angle HXM + \angle MXA) + \angle ACH \\ &= \left(180^\circ - \left(\frac{1}{2}\angle BAC + \angle ACB\right)\right) + \frac{1}{2}\angle BAC + \angle ACB \\ &= 180^\circ,\end{aligned}$$

and so $AXHC$ is cyclic. Therefore $\angle AXC = \angle AHC = 90^\circ$. $\square$

Solution 2. Let the circumcircle of $\triangle AMX$ meet AB again at A'. We have

$$\angle AA'M = \angle MXA = \angle DAC = \angle MAA',$$

Hence $\triangle AA'M$ is isosceles. Let M' be the midpoint of AA' and so $MM' \perp AB$. By the midpoint theorem, $MM' \parallel DA'$ which means that $DA' \perp AB$. Let H be the foot of the perpendicular from A to BC. Then H lies between B and D (because $AB < AC$). Since $DA' \perp AB$ we have that $\angle DA'A = \angle DHA = 90^\circ$ and so the quadrilateral $AA'HD$ is cyclic.

Observe that

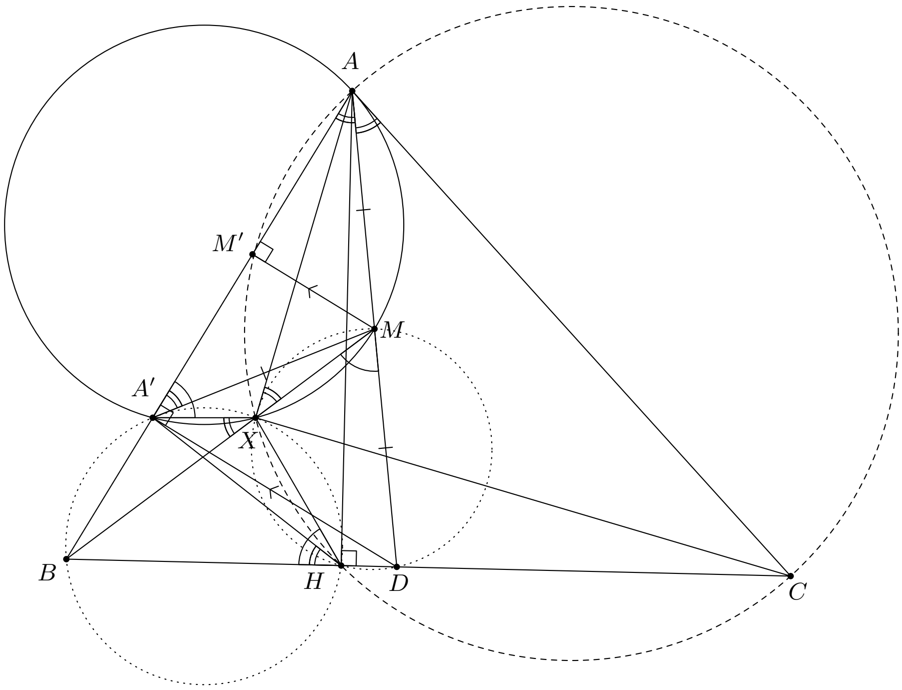
- $\angle BXA' = 180^\circ - \angle A'XM = \angle A'AD = 90^\circ - \angle A'DA = 90^\circ - \angle A'HA = \angle BHA'$ and so $A'BHX$ is cyclic;

- $\angle XMD = 180^\circ - \angle XMA = \angle AA'X = 180^\circ - \angle XA'B = \angle XHB$, and so $MXHD$ is cyclic (or you can simply invoke Miquel's Theorem applied to point X in $\triangle ABD$!)

Since $MXHD$ is cyclic, we have $\angle HXM = 180^\circ - \angle MDH$ and so

$$\begin{aligned}\angle AXH + \angle ACH &= (\angle HXM + \angle MXA) + \angle ACH \\ &= \left(180^\circ - \left(\frac{1}{2}\angle BAC + \angle ACB\right)\right) + \frac{1}{2}\angle BAC + \angle ACB \\ &= 180^\circ,\end{aligned}$$

and so $AXHC$ is cyclic. Therefore $\angle AXC = \angle AHC = 90^\circ$.



$\square$

Problem 66. Let $\lambda \geqslant 1$ be a real number and let n be a positive integer such that

$$\lfloor \lambda^{n+1} \rfloor, \lfloor \lambda^{n+2} \rfloor, \ldots, \lfloor \lambda^{4n} \rfloor \text{ are all perfect squares,}$$

where $\lfloor x \rfloor$ denotes the greatest integer not exceeding x.

Prove that $\lfloor \lambda \rfloor$ is a perfect square.

Solution. We will start with the case $n = 1$. Knowing that $\lfloor \lambda^2 \rfloor, \lfloor \lambda^3 \rfloor, \lfloor \lambda^4 \rfloor$ are perfect squares, we want to show that $\lfloor \lambda \rfloor$ is a perfect square. Let $a^2 = \lfloor \lambda^2 \rfloor$ for some positive integer a. Then

$$a^2 \leqslant \lambda^2 < a^2 + 1 < (a+1)^2$$

and so $a \leqslant \lambda < a + 1$ which means that $\lfloor \lambda \rfloor = a$. Squaring the previous inequalities, we find that

$$(a^2)^2 \leqslant \lambda^4 < (a^2+1)^2$$

and, since $(a^2)^2$ and $(a^2+1)^2$ are consecutive perfect squares, $\lfloor \lambda^4 \rfloor = (a^2)^2 = a^4$. It follows that $\lambda^4 - a^4 < 1$. Since $\lambda \geqslant 1$ we find that

$$\lambda^3 - a^3 = (\lambda - a)(\lambda^2 + \lambda a + a^2) \leqslant (\lambda - a)(\lambda(\lambda^2 + \lambda a + a^2) + a^3) = \lambda^4 - a^4$$

and so $\lfloor \lambda^3 \rfloor = a^3$, which is a perfect square by assumption. Hence $a = \lfloor \lambda \rfloor$ is a perfect square.

We will now show the general case by induction on n. We have established the base case $n = 1$. For the inductive step, suppose that if $\lfloor \lambda^{n+1} \rfloor, \lfloor \lambda^{n+2} \rfloor, \ldots, \lfloor \lambda^{4n} \rfloor$ are all perfect squares, then $\lfloor \lambda \rfloor$ is a perfect square. From the fact that $\lfloor \lambda^{(n+1)+1} \rfloor, \lfloor \lambda^{(n+1)+2} \rfloor, \ldots, \lfloor \lambda^{4(n+1)} \rfloor$ are all perfect squares we want to show that $\lfloor \lambda \rfloor$ is a perfect square. If we knew that $\lfloor \lambda^{n+1} \rfloor$ is a perfect square, we could apply the inductive hypothesis to complete the proof. Now let $\Lambda = \lambda^{n+1}$. Specifically, $\lfloor \Lambda^2 \rfloor, \lfloor \Lambda^3 \rfloor, \lfloor \Lambda^4 \rfloor$ are three of our perfect squares. Hence it follows that $\lfloor \lambda^{n+1} \rfloor = \lfloor \Lambda \rfloor$ is a perfect square. $\square$

Problem 67. Let x be a real number with $0 < x < 1$ and let $0.c_1c_2c_3\dots$ be the decimal expansion of x. Let $B(x)$ be the set of all substrings $c_ic_{i+1}\cdots c_{i+5}$, where $i = 1, 2, \dots$, in the decimal expansion of x that consist of 6 consecutive digits.
For instance, $B(1/22) = \{045454, 454545, 545454\}$.

Find the minimum number of elements of $B(x)$ as x varies over the irrational numbers between 0 and 1 (i.e. among all real numbers in that range whose decimal expansion is neither finite nor eventually periodic).

Solution. The answer is 7.

We will show that

(1) there exists an irrational x such that $B(x)$ has no more than seven elements, and that

(2) each real y such that $B(y)$ has six or fewer elements is rational.

For (1), consider the following number x: the first digit of the decimal expansion of x is 1, followed by five zeros, then another 1, followed by six zeros, followed by 1, followed by seven zeros, and so on, so that the decimal expansion of x is comprised of ones separated by strings of zeros of increasing length. Clearly

$$B(x) = \{000000, 000001, 000010, 000100, 001000, 010000, 100000\}\,.$$

Now we show that x is irrational. If x were rational, then, after a certain kth digit, its decimal expansion would be periodic. Let ℓ be the length of the *repetend* (technical name for a string of repeating decimal digits). Since the expansion of x contains infinite strings of zeros of lengths greater than 2ℓ, one of these must follow the kth digit and would contain a complete repetend. Therefore the repetend would have to contain ℓ zeros, which contradicts the fact that the extension of x contains infinite ones, and this proves (1).

Alternatively, we can prove (1) as follows: Let $d_i(x)$ be the ith digit after the decimal point of the number x between 0 and 1. Consider any irrational number α in this interval. For $i = 1, \dots, 6$ and $j = 0, \dots, 9$, we define $\alpha_{i,j}$ as follows

$$d_k(\alpha_{i,j}) = \begin{cases} 1 & \text{if } k = 6n + i \text{ for some } n \text{ and } d_k(\alpha) > j, \\ 0 & \text{otherwise.} \end{cases}$$

Adding these numbers, we immediately see that, of the kth digits, exactly $d_k(\alpha)$ are ones and the others zeros. Therefore α is the sum of the $\alpha_{i,j}$. Consequently, at least one of these numbers, which we will call x, is irrational. On the other hand, the digits of all $\alpha_{i,j}$ are only zeros and ones, and there is at most one 1 every six consecutive digits. Therefore

$$B(x) \subseteq \{000000, 000001, 000010, 000100, 001000, 010000, 100000\}$$

For (2), let $B_k(y)$ be the set of substrings of length k of the decimal expansion of y. We will show by induction on k that if $B_k(y)$ has k or fewer elements, then y is rational. This would clearly prove (2) as $B(y) = B_6(y)$.

The case with $k = 1$ is obvious. We will assume that the property holds for k and show that it holds for $k + 1$. We will base our proof on the fact that each substring of the digits of y of length $k + 1$ begins with a substring, which we will call a corresponding substring, of length k. Conversely, given a substring of length k, it must correspond to at least one substring of length $k + 1$. In particular, $B_k(y)$ has at most as many elements as $B_{k+1}(y)$. Given a y, we will study the number of elements of $B_k(y)$. If this number is less than or equal to k, the the result follows from the case k. On the other hand, the number of elements of $B_k(y)$ cannot exceed $k + 1$, otherwise $B_{k+1}(y)$ would also have more than $k + 1$ elements, contradicting the hypothesis. We can therefore assume that both $B_k(y)$ and $B_{k+1}(y)$ have exactly $k + 1$ elements. The correspondence described above is therefore one-to-one. In other words, each digit after the kth of the expansion of y is uniquely determined by the previous k digits. Since there are $k+1$ possible substrings of k digits, there must exist i and j such that $i < j$ and the k digits of the decimal expansion of y after the ith digit coincide with the k digits after the jth digit. But then the whole expansion after the ith digit must coincide with the expansion after the jth digit. This means that the digits between the ith digit exclusive and the jth digit inclusive repeat periodically, therefore y is rational, and this proves (2). $\square$

Problem 68. In a certain club, some pairs of members are friends. Given $k \geqslant 3$, we say that a club is *k-good* if every group of k members can be seated around a round table such that every two neighbours are friends.

(a) Prove that if a club is 6-good then it is 7-good.
(b) Give an example of a club which is 9-good but not 10-good.

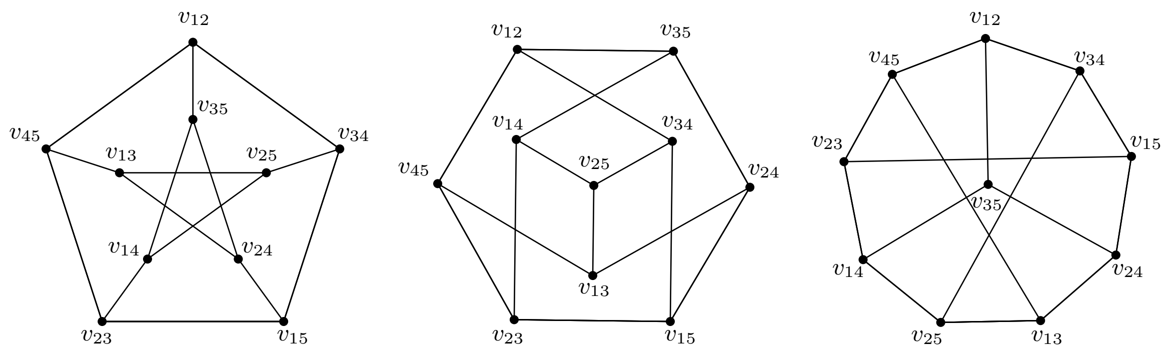
Solution. (a) Consider a 6-good club and denote some seven of its members by $A_1, \ldots, A_7$. It suffices to show that $A_1, \ldots, A_7$ can be seated around a table as required. Consider only friendships among $A_1, \ldots, A_7$. First, we show that every member has at least three friends.

Without loss of generality consider A_7. By assumption, $A_1, \ldots, A_6$ can be seated as required, hence A_7 has at least two friends. Without loss of generality, A_6 is one of them. By assumption, $A_1, \ldots, A_5, A_7$ can be seated as required, hence A_7 has at least two more friends apart from A_6 for a total of at least three friends.

Since every member has at least three friends, there exists a member with at least four friends (otherwise the number of friendly pairs equals $\frac{1}{2} \times 7 \times 3$, which is clearly impossible). Without loss of generality, assume A_7 has at least four friends.
By assumption, $A_2, \ldots, A_6$ can be seated as required. In such a seating, some two of the four friends of A_7 are neighbours and we can seat A_7 in between them.

(b) The problem is equivalent to finding an example of a (connected) graph with 10 vertices that has no Hamiltonian cycle but upon deleting one vertex the graph has a Hamiltonian cycle. (Recall that a *Hamiltonian path* is a path in an undirected or directed graph that visits each vertex exactly once and a *Hamiltonian cycle* is a Hamiltonian path that is a cycle.)

The example we seek is given by the (famous) *Petersen graph* with 10 vertices and 15 edges. This is a graph whose vertices correspond to the 2-element subsets of a 5-element set and whose edges correspond to the pairs of disjoint 2-element subsets. Below the graph is drawn in three different ways where the vertices are labelled as v_{ij}, with ij corresponding to the 2-element subset $\{i, j\}$ of the 5-element set $\{1, 2, 3, 4, 5\}$. Note that for each 2-element subset of a 5-element set, there exactly three ways to choose a 2-element set from the remaining 3 elements, so every vertex of the graph has degree 3.



To see that the Petersen graph has no Hamiltonian cycle, consider the edges in the cut disconnecting the inner 5-cycle from the outer one. If there is a Hamiltonian cycle, an even number of these edges must be chosen. If only two of them are chosen, their end-vertices must be adjacent in the two 5-cycles, which is not possible. Hence 4 of them are chosen. Assume that the top edge of the cut is not chosen (all the other cases are the same by symmetry). Of the 5 edges in the outer cycle, the two top edges must be chosen, the two side edges must not be chosen, and hence the bottom edge must be chosen. The top two edges in the inner cycle must be chosen, but this completes a non-spanning cycle, which cannot be part of a Hamiltonian cycle. Finally note that by deleting one vertex from the Petersen graph, we have a Hamiltonian cycle on 9 vertices (which can easily be seen on the third figure on the right!). $\square$